

TIC 330901906

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	3.732068 d	VSX	no match
Epoch (t0)	1740.9150 BTJD	ASAS-SN	no match
Duration	0.2440 d	Gaia eclipsing-binary table	no match
Depth	0.498 % (4,983 ppm)	Gaia variable classification	no match
SNR	60.1	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	29 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.19 (gate >= 0.50)		
Robust depth sigma	58.6 (gate >= 7.0)		
Odd/even z-score	0.36 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

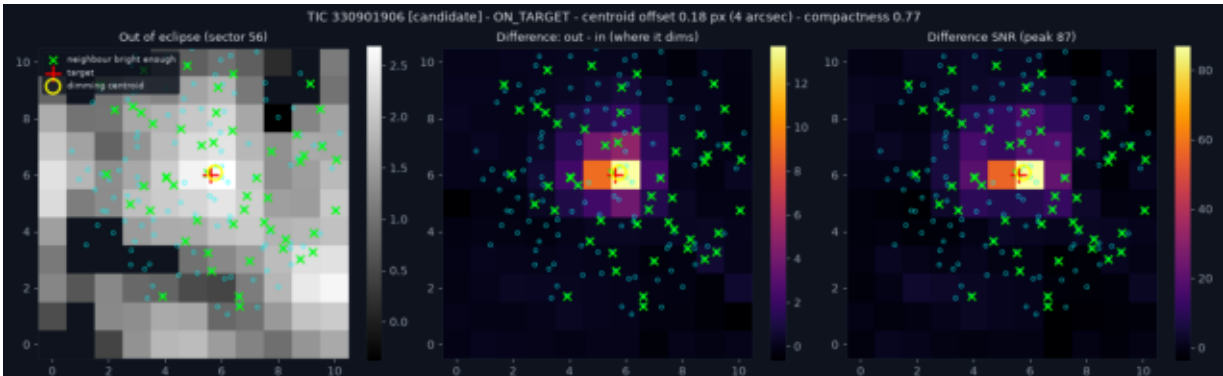
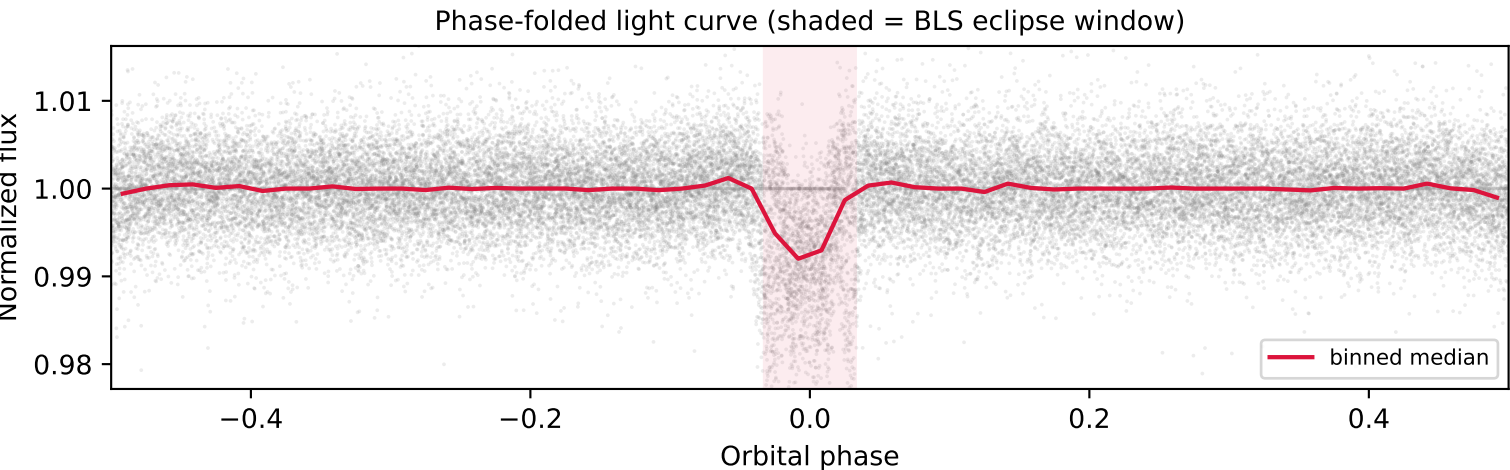
Sector 56 light-curve dip in this sector: 38.2 sigma t0 refined by 0.0075 phase

Difference-image peak SNR 87.24 (gate >= 5.0) Compactness 0.77 (gate >= 0.50)

Dimming centroid offset 0.18 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 2009125529

0.25 px away, dTmag +4.30 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.066 px (dimming centroid 0.182 px from target vs 0.248 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.